

11. Generalized Regression Neural Network (GRNN)

Algorithm implemented: Specht's GRNN (Nadaraya-Watson kernel regression) from scratch.

Dataset: real motorcycle accelerometer data (`data/motorcycle.csv` , R `MASS::mcycle`): time after impact vs head acceleration.

```
# ---- Core Libraries ----
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.base import BaseEstimator, RegressorMixin
from sklearn.model_selection import KFold, train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import r2_score, root_mean_squared_error

# Styling setup
plt.style.use('seaborn-v0_8-whitegrid' if 'seaborn-v0_8-whitegrid' in plt.style.available else 'default')
plt.rcParams['font.sans-serif'] = 'DejaVu Sans'
plt.rcParams['figure.figsize'] = (10, 6)
plt.rcParams['figure.dpi'] = 110
np.random.seed(42)
```

GRNN Architecture

Step-by-Step Algorithm (Generalized Regression Neural Network)

1. **Input layer:** receive the d-dimensional query vector x and pass it to every pattern neuron.
2. **Pattern layer:** with N training points, each unit i computes a Gaussian activation $p_i(x) = \exp(-\|x - x_i\|^2 / (2\sigma^2))$.
3. **Summation layer:** compute two sums over all pattern units:
 - $S(x) = \sum_i y_i \cdot p_i(x)$ (weighted by training targets)
 - $D(x) = \sum_i p_i(x)$ (normalization)
4. **Output layer:** predict the conditional mean $\hat{y}(x) = S(x) / D(x)$ (Nadaraya-Watson estimator).
5. **Training:** one pass - just store the training set; **no** backpropagation or iterative optimization.
6. **Tuning:** only σ must be chosen (small σ = memorizes noise; large σ = over-smooth).

```
class GRNN(BaseEstimator, RegressorMixin):
    """
    Generalized Regression Neural Network (Specht, 1991) - vectorized implementation.

    Prediction is the Nadaraya-Watson kernel regression estimate:
        y_hat(x) = sum_i y_i * exp(-||x - x_i||^2 / (2*sigma^2))
                  / sum_i      exp(-||x - x_i||^2 / (2*sigma^2))

    Training is one-pass: the model simply stores the training set (no gradient descent).
    The only hyperparameter is the smoothing spread sigma.
    """

    def __init__(self, sigma=1.0):
        self.sigma = sigma

    def fit(self, X, y):
        # "Training" = memorizing the data (pattern + summation layers)
        self.X_train_ = np.asarray(X, dtype=np.float64)
        self.y_train_ = np.asarray(y, dtype=np.float64).ravel()
        return self

    def predict(self, X):
        X_test = np.asarray(X, dtype=np.float64)

        # ---- Pattern Layer: squared Euclidean distance between each test and train point ----
        # Uses the expansion ||a-b||^2 = ||a||^2 + ||b||^2 - 2 a.b for efficiency.
        dist_sq = np.sum(X_test**2, axis=1, keepdims=True) + \
```

```

        np.sum(self.X_train**2, axis=1, keepdims=True).T - \
        2 * np.dot(X_test, self.X_train.T)
    dist_sq = np.maximum(dist_sq, 0.0) # guard against tiny negative values from rounding

    # ---- Pattern Layer activation: Gaussian kernel of each training point ----
    kernels = np.exp(-dist_sq / (2.0 * (self.sigma ** 2)))

    # ---- Summation Layer: D-sum (denominator) and S-sum (numerator) ----
    D = np.sum(kernels, axis=1) # unweighted sum of activations
    S = np.dot(kernels, self.y_train_) # activation-weighted sum of targets

    # ---- Output Layer: normalized weighted average (prevents 0/0 for isolated queries) ----
    D_safe = np.where(D < 1e-12, 1e-12, D)
    return S / D_safe

```

```
print("Custom vectorized GRNN class defined!")
```

Custom vectorized GRNN class defined!

Smoothing Parameter σ

The Role of the Smoothing Parameter σ

1. Fit GRNNs with a small, medium and large σ on the same noisy 1D data.
2. Small σ (0.05): each training point dominates its own neighborhood -> wiggly curve, high variance.
3. Large σ (1.50): all points influence every query -> flat curve, high bias.
4. Medium σ (≈ 0.35): smooth interpolation that follows the true underlying function.
5. Compare training R^2 and curve shapes to see the bias-variance trade-off.

Hyperparameter Optimization Process (5-Fold CV)

1. Define a grid of candidate σ values from 0.05 to 1.0.
2. Split the data into 5 fixed folds.
3. For each σ : fit on 4 folds, compute RMSE on the held-out fold, then average the 5 scores.
4. Select the σ with the lowest average CV RMSE.
5. Plot the CV error curve to confirm a clear optimum.

```

# ---- Load the real motorcycle impact data (MASS::mcycle) ----
moto = pd.read_csv('../data/motorcycle.csv')
X = StandardScaler().fit_transform(moto['times'].values.reshape(-1, 1)) # time after impact (scaled)
y = moto['accel'].values # head acceleration (g)
print(f"Motorcycle data: {len(moto)} observations | time {moto.times.min()}-{moto.times.max()} ms | accel {moto.accel.min():.0f} to {moto.accel.max():.0f} g")

X_eval = np.linspace(X.min(), X.max(), 300).reshape(-1, 1)

# Under-smoothing, near-optimal, over-smoothing
sigmas = [0.05, 0.25, 1.0]
colors = ['purple', 'crimson', 'teal']

fig, axes = plt.subplots(1, 3, figsize=(18, 5))

for ax, sig, col in zip(axes, sigmas, colors):
    # Fit a fresh GRNN for each sigma (training is instant)
    grnn_demo = GRNN(sigma=sig).fit(X, y)
    y_pred_demo = grnn_demo.predict(X_eval)

    ax.scatter(X, y, color='navy', s=25, alpha=0.6, label='Observations')
    ax.plot(X_eval, y_pred_demo, color=col, lw=2.5, label=f'GRNN ( $\sigma={sig}$ )')

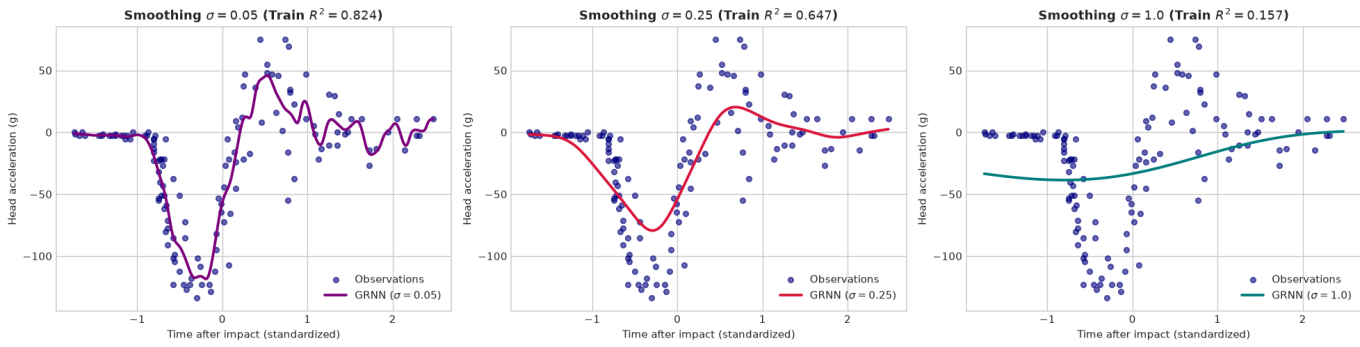
    # Training R^2 shows the bias-variance behavior for each sigma
    r2 = r2_score(y, grnn_demo.predict(X))
    ax.set_title(f"Smoothing  $\sigma={sig}$  (Train  $R^2 = {r2:.3f}$ )", fontweight='bold')
    ax.set_xlabel("Time after impact (standardized)")
    ax.set_ylabel("Head acceleration (g)")
    ax.legend(loc='lower right')

plt.suptitle("GRNN Behavior on Real Motorcycle Impact Data ( $\sigma$  sweep)", fontsize=15, fontweight='bold')
plt.tight_layout()
plt.show()

```

Motorcycle data: 133 observations | time 2.4-57.6 ms | accel -134 to 75 g

GRNN Behavior on Real Motorcycle Impact Data (σ sweep)



```
# ---- 5-fold CV grid search for the optimal smoothing sigma ----
sigma_candidates = np.linspace(0.02, 1.0, 50)
cv_scores = []
kf = KFold(n_splits=5, shuffle=True, random_state=42) # fixed folds for a fair comparison

for sig in sigma_candidates:
    fold_rmse = []
    for train_idx, val_idx in kf.split(X):
        # Split the data into this fold's train/validation parts
        X_tr, y_tr = X[train_idx], y[train_idx]
        X_va, y_va = X[val_idx], y[val_idx]

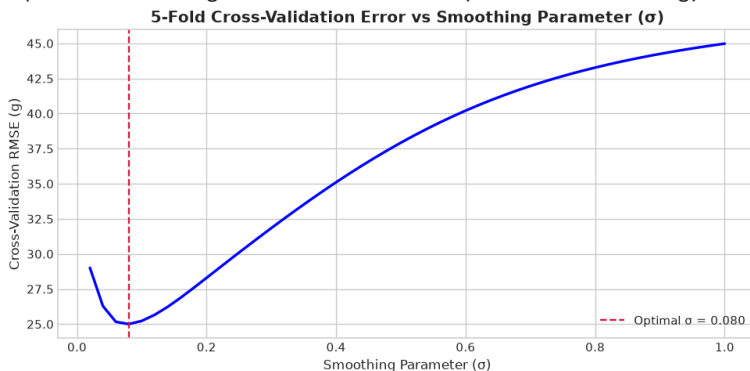
        # Fit GRNN on the training fold and score on the validation fold
        model = GRNN(sigma=sig).fit(X_tr, y_tr)
        preds = model.predict(X_va)
        fold_rmse.append(root_mean_squared_error(y_va, preds))

    cv_scores.append(np.mean(fold_rmse)) # average CV error for this sigma

best_idx = np.argmin(cv_scores)
best_sigma = sigma_candidates[best_idx]
print(f"Optimal Smoothing Parameter  $\sigma$ : {best_sigma:.3f} (CV RMSE: {cv_scores[best_idx]:.4f} g)")

# ---- Plot the CV error curve and mark the optimum ----
plt.figure(figsize=(9, 4.5))
plt.plot(sigma_candidates, cv_scores, 'b-', lw=2.2)
plt.axvline(x=best_sigma, color='crimson', linestyle='--', label=f'Optimal  $\sigma$  = {best_sigma:.3f}')
plt.title("5-Fold Cross-Validation Error vs Smoothing Parameter ( $\sigma$ )", fontsize=13, fontweight='bold')
plt.xlabel("Smoothing Parameter ( $\sigma$ )", fontsize=11)
plt.ylabel("Cross-Validation RMSE (g)", fontsize=11)
plt.legend()
plt.tight_layout()
plt.show()
```

Optimal Smoothing Parameter σ : 0.080 (CV RMSE: 25.0023 g)



```
# ---- Final model: refit with the CV-selected sigma and evaluate on a held-out split ----
X_tr, X_te, y_tr, y_te = train_test_split(X, y, test_size=0.25, random_state=42)
final_model = GRNN(sigma=best_sigma).fit(X_tr, y_tr)
y_hat = final_model.predict(X_te)

print(f"Final GRNN with CV-selected sigma = {best_sigma:.3f}")
print(f" - Train R2: {r2_score(y_tr, final_model.predict(X_tr)):.4f}")
print(f" - Test R2: {r2_score(y_te, y_hat):.4f}")
print(f" - Test RMSE: {root_mean_squared_error(y_te, y_hat):.4f} g")
```

Final GRNN with CV-selected sigma = 0.080

- Train R²: 0.8073
- Test R²: 0.7255
- Test RMSE: 22.7760 g